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PERSONAL INFORMATION

# Dr. Jagrati Talreja, PhD




1015 Dewey Street, Greensboro, North Carolina, USA, 27405.

+1 336 594 9171 +91 790 5426 423

jtalreja@ncat.edu, talrejajagrati02@gmail.com

https://jagratitalreja01.github.io/portfolio/

https://github.com/JagratiTalreja01

https://www.linkedin.com/in/dr-jagrati-talreja-phd-113054136


Date of birth 22 September 1996 | Nationality Indian

RESEARCH INTEREST

Data Science, Artificial Intelligence, Computer Vision, Deep Learning, Machine Learning, Geospatial Data Analyst (Sentinel-1 & Sentinel-2), Image and Video Processing, Robotics, XAI, Remote Sensing and Satellite Image Analysis.

Jan 2025 – Current

## Postdoctoral Fellow, Co-Lead & Co-PI for NASA and NSF Funded Projects

**North Carolina A&T State University, North Carolina, USA**

**Sponsored by NASA and NSF**

As a Postdoctoral Fellow and Co-Lead/Co-PI on federally funded projects, I lead and conduct research on AI and quantum-inspired methods for satellite-based environmental monitoring.

**Research Leadership:** Co-leading project design and implementation of AI-driven methodologies for remote sensing using satellite imagery.

**AI Model Development and Remote Sensing:** Developing CNN, GAN, and Transformer-based frameworks for satellite image processing, SAR–optical translation, multimodal data fusion, and flood monitoring.

**Collaboration and Mentorship:** Working with NASA collaborators and interdisciplinary teams while mentoring graduate and undergraduate researchers.

**Quantum Computing Initiative:** Initiating research on quantum-inspired machine learning for remote sensing and satellite data analytics.

**Research Dissemination:** Building scalable geospatial data pipelines and publishing results in peer-reviewed journals.

Jan 2020 – 2024

## Project Work, PhD

Department of Electrical Engineering, Faculty of Engineering
Chulalongkorn University, Bangkok, Thailand

- **(1.) Research Guide:** Assoc. Prof. Dr. Supavadee Aramvith and Dr. Watchara Ruangsang.
- **Project**: Real-time image super-resolution reconstruction for system-on-chip FPGA
- **Project Description**: The development of a Deep Learning algorithm for SoC FPGA that can be employed for image super-resolution in real time for fast processing of cutting-edge devices.
- **Skills Acquired**: Deep Learning, CUDA, CuDNN, TQDM, FPGA, working with SoC & PyTorch.
- **(2.) Research Guide**: Asst. Prof. Dr. Charnchai Pluempitiwiriyawej.
- **Project**: Offline Signature Verification using shape Dissimilarities
- **Project Description**: This project aims to create an offline signature verification system using shape dissimilarities. It authenticates signatures from scanned images without requiring specialised hardware, comparing signature shapes and features to determine their similarity or dissimilarity, providing a reliable authentication method.
- **Skills Acquired**: MATLAB & Latex.

- **(3.) Research Guide**: Asst. Prof. Dr. Suree Pumrin.
- **Project**: Contour Detection and color tracking using Raspberry Pi and OpenCV Python
- **Project Description**: This project aims to develop a system using Raspberry Pi and OpenCV in Python for contour detection and color tracking. Contour detection is the process of identifying the boundaries of objects in an image, while color tracking involves locating and following specific colors within the image.
- **Skills Acquired**: Raspberry Pi, OpenCV & Python.

August 2020 – 2024

## Teaching Assistant, PhD

Department of Electrical Engineering, Faculty of Engineering
Chulalongkorn University, Bangkok, Thailand

As a Teaching Assistant, my role typically encompasses several key responsibilities:

- **Curriculum Design & Development**: Developed a cutting-edge Electrical Engineering curriculum with comprehensive materials for Computer Vision and Digital Signal Processing.
- **Lecture Delivery & Lab Management**: Conducted lectures and labs on Image and Video Processing, integrating theory with practical applications using specialised tools.
- **Student Evaluation & Feedback**: Implemented diverse assessments, including assignments, projects, and exams, and provided constructive feedback to enhance student learning.
- **Research & Professional Development**: Pursued research and professional development by attending conferences, and seminars, and engaging in collaborative projects to stay at the forefront of technological advancements.
- **Skills Acquired**: Mentoring and Support, Teamwork and Collaboration & Adaptability.

June 2018 – June 2019

## Research Assistant / Internship

Department of Electrical Engineering, Faculty of Engineering
Indian Institute of Technology (IIT), Kanpur, India

- **Research Guide**: Prof. Dr. K.S. Venkatesh.
- **Project**: Disruption detection (Digital Image Processing using MATLAB)
- **Project Description**: The development of a software application namely Disruption Detector to detect the motion of an object based on its image depicting the disruption of an object for security purposes.
- **Skills Acquired**: Image Acquisition Tools, MATLAB & GitHub.

Jan 2018 – Dec 2018

## Project Work

Department of Electronics & Communication Engineering, Faculty of Engineering
Pranveer Singh Institute of Technology, Uttar Pradesh, India.

- **(1. ) Research Guide**: Dr. Raghvendra Singh, Dr. Vivek Kumar, Dr. Yogesh Mohan Dubey.
- **Project**: Medi-Cart: Telemetry using RFID
- **Project Description**: The development/designing of RFID-Bluetooth-based medical/health care device that can be employed for automating and streamlining patient identification and medical records process in hospitals for health care management systems.
- **Skills Acquired**: C++ & RFID.
- **(2. )Development of Voice Control Robo Car**
  - **Skills Acquired**: Arduino, Microcontrollers (Arduino Atmega 2560 & Minitron V1.0), Binary program burning in hardware using XML Java.
- **(3. )School Management System Design Using Java**
  - **Skills Acquired**: Java (NetBeans IDE) & DBMS (MySQL).


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## EDUCATION AND TRAINING

2019–2024

### PhD (5-Year Integrated) in Data Science & AI

Multimedia Data Analytics and Processing Unit (MDAP)
Department of Electrical Engineering, Faculty of Engineering
Chulalongkorn University, Bangkok, Thailand
**Thesis:** Single Image Super-Resolution using Deep Attention Networks.
**Achievements:-**

- Received THE SECOND CENTURY FUND (C2F) Chulalongkorn University scholarship for PhD students 1st batch, 2019, recognising high efficiency.
- Awarded the Overseas Research Experience Scholarship for Graduate Students, Fiscal Year 2024.
- Honoured with the 90th Anniversary of Chulalongkorn University Scholarship under the Ratchadapisek Somphot Endowment Fund, receiving a Lenovo Legion 7i Gen 7 (16", 2023) & iPad Air 6 2024.

2015–2019

### Bachelor of Technology (B.Tech) in Electronic & Communication Engineering

Department of Electrical Engineering, Faculty of Engineering, Dr. A. P. J. Abdul Kalam Technical University (AKTU), Lucknow, India. **Project:** Medi-Cart: Telemetry using RFID.
**Achievements:-**

- Gold Medal for excellence in Institutional Academia scoring 80.93%.
- Got Patent for the final year project.
- Got Selected for the top 100 Engineering & Science projects in Uttar-Pradesh (2019).
- Awarded with University Scholarship for 4 Years (25% off Tuition Fees)
- Awarded with the State Government Scholarship for 4 Years (65% off Tuition Fee )

## CONFERENCES & SEMINARS

- **(**Speaker) International Geoscience and Remote Sensing Symposium (IGARSS), August 10-15, 2026.
- **(**Speaker) American Society for Photogrammetry and Remote Sensing (ASPRS), February 15-19, 2026.
- **(**Speaker) American Geophysical Union (AGU), December 15-19, 2025.
- **(**Speaker) Geographic Information Systems (GIS) Day, November 19, 2025.
- **(**Host) IEEE CASS Day 2022 Thailand Chapter, December 6, 2023.
- **(**Host) R10 Robotics Competition 2022 for Healthcare IEEE Region 10 Thailand Section. Chulalongkorn University, December 18-19, 2022.
- **(**Host) Robotics Competition 2023 for Managing Climate Change for a Better World, IEEE Region 10 Thailand Section. Chulalongkorn University, December 9-11, 2023.

## PATENT

**1. Medi-Cart: Telemetry using RFID Indian patent number: 201811037266A, dated: 19/10/2018**

**Inventors**: Jatin Vasisth, Km Nandani, Harshita Saxena, **Jagrati Talreja**, Raghvendra Singh, Vivek Kumar, Yogesh Mohan Dubey.

**2. SafePath Navigate to Safety — US Application Number 63/929,945, dated: 12/03/2025**

**Inventors**: Tewodros Syum Gebre, **Jagrati Talreja**, Leila Hashemi-Beni.


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## PUBLICATIONS

**Published and Accepted Articles:-**

[1] **J. Talreja**, T.S. Gebre, L.H. Beni, "FLoRA: Fusion-Latent for Optical Reconstruction and Flood Area Segmentation via Cross-Modal Multi-Task Distillation Network." (**IEEE Transactions on Geoscience & Remote Sensing**), DOI: 10.1109/TGRS.2026.3701300

[2] **J. Talreja**, T.S. Gebre, L.H. Beni, "DPolTransGAN: Scattering Guided Hybrid Transformer-GAN for SAR-Optical Image Translation using Dual Polarization SAR." (**IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing**), DOI: 10.1109/JSTARS.2026.3691677

[3] **J. Talreja**, S. Aramvith and T. Onoye, "DANS- Deep Attention Network for Single-Image Super-Resolution," in **IEEE Access**, DOI: 10.1109/ACCESS.2023.3302692

[4] **J. Talreja**, S. Aramvith and T. Onoye "Deep Hybrid Transformer Network for Single-Image Super-Resolution (DHTCUN)," in **IEEE Access**, DOI: 10.1109/ACCESS.2024.3450300

[5] **J. Talreja**, S. Aramvith and T. Onoye "Xception-Based Transformer Network for Single-Image Super Resolution " **Springer Complex & Intelligent Systems** , DOI: https://doi.org/10.1007/s40747-024-01760-1

[6] **J. Talreja**, D. Chauhan, "Advanced Computational Techniques: Bridging Metaheuristic Optimization and Deep Learning for Material Design through Image Enhancement", **Elsevier Series in Mechanics of Advanced Materials**, DOI: https://doi.org/10.1016/B978-0-443-29162-3.00007-1.

[7] **J. Talreja**, LH Beni, "SARGANet: A GAN-U-Net Hybrid Framework for Super-Resolution of Sentinel-1 SAR Imagery", **Earth Science Informatics**, DOI: https://doi.org/10.1007/s12145-026-02164-0

[8] **J. Talreja**, LH Beni, "STCN-SR: A Spatial Transformer-CNN Framework for Enhancing Sentinel-2 Imagery, **Geomatica**, DOI: 10.1016/j.geomat.2026.100135

[9] T.S. Gebre, **J. Talreja**, L.H. Beni "Advanced Flood Prediction with Physics-Guided Deep Learning: Combining UNet, FNO, and SAR/Optical Imagery" in **IEEE Radar Conference 2026** (**Accepted**)

[10] T.S. Gebre, **J. Talreja** L.H. Beni, "Multi-Modal Attention for Automated Disaster Damage Assessment Using Remote Sensing Imagery and Deep Learning" International Society for Photogrammetry and Remote Sensing **(ISPRS 2026)** (**Accepted**)

[11] **J. Talreja**, M. Gupta, "A Survey on Motion Estimation Techniques," Book on "Digital Signal Processing, Computer Vision and Image Processing," in **Wiley- Scrivener Publisher (John Wiley & Sons), (2024)**, (**Accepted**)

[12] **J. Talreja**, M. Gupta, "Combined Edge Detection using Wavelet Transform and Signal Readjustment: A Survey," Book on "Digital Signal Processing, Computer Vision and Image Processing," in **Wiley- Scrivener Publisher (John Wiley & Sons), (2024)**, (**Accepted**)

[13] **J. Talreja**, T.S. Gebre, L.H. Beni, "Cross-Polarization Fusion of VV and VH SAR observations for improved flood mapping." in IEEE International Geoscience and Remote Sensing Symposium **(IGARSS 2026)** (**Accepted**)

[14] T.S. Gebre, **J. Talreja** L.H. Beni, "Physics-Informed Machine Learning for Short-Term Flood Predection" in IEEE International Geoscience and Remote Sensing Symposium **(IGARSS 2026)** (**Accepted**)

[15] **J. Talreja**, LH Beni, D. Chauhan, M. Ashfaq, N. Talreja, H. Singh, "Deep Learning-Based Image Enhancement for Material Imagery: A Comprehensive Review" (**Accepted**)

[16] **J. Talreja**, T.S. Gebre, L.H. Beni, "Spatially-Aware Quantum Encoding for Flood Patch Classication in Multi-Modal Remote Sensing." (**Under-Review**)


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## ACHIEVEMENTS

- **GRE 304(2020).**
- **IELTS 6.5 (2019).**
- **JEE MAINS Qualified (2015).**

## TECHNICAL SKILLS

**Data Science**: Data Processing, Geospatial Data Analysis(Google Earth Engine and Alaska Satellite Facility), Computer Vision, Machine/Deep Learning, Data Visualisation, NLP.
**Programming**: Python (PyTorch, Tensorflow, OpenCV, Keras, Scikit-learn, Numpy, Pandas, Matplotlib), C and C++, MATLAB
**Techniques**: Neural Networks, Transformer Networks, Generative Adversarial Networks, Diffusion Models, FPGA, Autoencoder, Explainable AI, Transfer Learning, Segmentation, Regression Analysis, Classification, Clustering, Attention Mechanisms, Digital Image Processing, and so on.
**Software and OS**: ArcGIS Pro, SNAP, VScode, PyCharm, Notepad++, Spyder, Latex, Visual Studio, MS Visio, Office 365, Windows, Linux, WSL (Windows Subsystem Linux).
**Hardware**: Experience with GPU-based model training on NVIDIA A4000, A6000, and Lambda GPU clusters (A100/H100). NVIDIA GeForce GTX Series GPUs, Microcontrollers (Arduino, Raspberry PI), SoC FPGA.
**Others**: CUDA, CuDNN, TQDM, Anaconda, Colab, Git, Overleaf, Jupyter Notebook, Linux OS, NetBeans IDE, MySQL, Eclipse IDE, Android Studio.
**Programming Languages**: HTML, CSS, JAVASCRIPT, SQL, NoSQL, Java, C, Node.js, Bootstrap.
**Misc.**: Academic research, teaching, training, consultation, LATEX typesetting and publishing.

## INTERNATIONAL CONFERENCES AND JOURNALS

- **Reviewer for IEEE Access.**
- **Reviewer for PeerJ.**
- **Reviewer for Springer.**
- **Reviewer for IEEE Signal Processing Society.**
- **Reviewer for IEMECON, 12th International Conference on the Internet of Everything, Microwave, Embedded, Communication, and Networks.**

## PERSONAL SKILLS

Other languages

| | UNDERSTANDING | | SPEAKING | | WRITING |
|---|---|---|---|---|---|
| | Listening | Reading | Spoken interaction | Spoken production | |
| English **IELTS (Academic)=6.5 Overall** | 6.5 | 6.5 | 6.5 | 6.5 | 6.5 |

Driving licence: India (Two & Four Wheeler), Thailand (Two & Four Wheeler), USA (Four Wheeler))


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## CERTIFICATIONS & COURSES

- **Certificate for participation in Robotics Competition 2023 for Managing Climate Change for a Better World, 2023.**
- **Certificate for participation in IEEE CASS Day 2022 Thailand Chapter, 2022.**
- **Supervised Machine Learning: Regression and Classification**- Offered by Stanford University and Deep learning, AI.
- **Certificate in the development of Voice Control Robo Car**, sponsored by i-3 Indya Technologies.
- **Certificate in the development of Aircraft**, sponsored by SkyFi Labs.
- **Certificate of Workshop in Ethical Hacking (Indian Institute of Technology Kanpur, India) sponsored by KYRION**.
- **Mathematics for Machine Learning Specialization** - Offered by Imperial College London
- **Certificate of Workshop in Cyber Security** sponsored by A.K.T.U and Uttar-Pradesh Police.
- **Computer Vision on Raspberry Pi - Beginner to Advanced** - Udemy Project
- **Learn the Fundamentals of VHDL and FPGA Development** - Udemy Project
- **PyTorch for Deep Learning with Python Bootcamp** - Udemy Project
- **Explainable AI: Scene Classification and GradCam Visualization**- Coursera Project.
- **Convolutional Neural Networks in Python: CNN Computer Vision** - Udemy Project
- **Deep Learning with PyTorch** - Udemy Project
- **The Complete 2024 Web Development Bootcamp** - Udemy Project
- **The Complete Python Bootcamp** - Udemy Project
- **Selenium WebDriver with Java Advanced Frameworks** - Udemy Project
- **Image Super Resolution Using Autoencoders in Keras**- Coursera Project.

## ACADEMIC REFERENCE

**Dr. Tewodros Syum Gebre**
Postdoctoral Research Associate
North Carolina A&T Technical State University, North Carolina, United States of America.
tsgebre@aggies.ncat.edu

**Dr. Muhammad Ashfaq**
Associate Professor
Chandigarh University, Punjab, India.
mohdashfaq007@gmail.com

**Dr. Divya Chauhan**
Water Quality Expert
Ministry of Jal Shakti, New Delhi, India.
deep2424@gmail.com


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